Michalis Niarchos

Systems Software Engineer
Low-Level, Networking, Linux Kernel

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Experience

January 2024 - May 2026

Ryzome, *Senior Software Engineer* - Singapore, SG (Remote)

Security and digital forensics solution, monitoring virtual machines at the hypervisor level.

- Contributed to the development of the kernel-side part of the solution (module, livepatch, kprobes).
- Redesigned the C userspace library providing communication interfaces between the userspace application and the kernel module (ioctl, UNIX sockets).
- Developed a kernel symbol resolution service in Go.
- Designed and implemented a custom live testing platform using Bash.
- Implemented VM instantiation tools in Python and Bash.
- Performed DevOps tasks, including CI/CD pipeline setup and maintenance, build automation, and infrastructure support.

https://www.ryzome.com

October 2022 - June 2023

Mobileum, *Senior Software Engineer, Contractor* - Singapore, SG (Remote)

Designed, implemented, tested and documented a complex customer-specific feature involving the DPI engine and the live analytics tool. The solution addressed the correlation of Control and User Plane traffic distributed across multiple DPI probes at different sites by the customer's network architecture and load balancing scheme.

- Analysed Control Plane and User Plane traffic.
- Designed and implemented a distributed messaging system enabling the master probe to collect Control Plane session information and synchronise it with slave probes for User Plane traffic correlation.
- Integrated the solution with the live analytics platform for unified processing and reporting across all probes.

https://www.mobileum.com/

February 2020 - May 2022

Niometrics, *Senior Software Engineer, Team Lead* - Athens, GR (Remote)

Led Control Plane development on the DPI engine, the core component of the real-time deep network analytics platform for telecommunications operators.

- Owned multiple core components of the high-performance DPI engine written in C, including MAP/CAP, RANAP, Radius, Diameter, PFCP, NAS, NGAP and location tracking.
- Designed and implemented new functionality based on PCAP analysis and 3GPP/ETSI specifications documentations.
- Acted as Control Plane team leader, coordinating development and technical direction.
- Contributed to feature feasibility assessment and effort estimation.
- Conducted code reviews to ensure code quality and maintainability.

https://www.niometrics.com/

December 2017 - January 2020

Niometrics, *Software Engineer* - Athens, GR (Remote)

Worked as part of the Control Plane team of the DPI engine.

- Contributed to the development and maintenance of the DPI engine, a large codebase tool written in C.
- Diagnosed and resolved functional and concurrency issues in the DPI engine.
- Investigated and fixed complex multithreading issues, such as AddressSanitizer interactions with real-time threads (https://groups.google.com/g/llvm-dev/c/-4ap4i8Dckk).
- Implemented performance optimizations.

https://www.niometrics.com/

January 2016 - November 2017

Information Technologies Institute, *Research Assistant* - Thessaloniki, GR

EU Funded Projects (Horizon 2020):
FORENSOR

- Ported computer vision algorithms to C for a low-memory, low-power multicore DSP on a non-commercial SoC developed by STMicroelectronics.
- Deployed, validated and tested on prototype hardware.

https://cordis.europa.eu/project/id/653355

MaTHiSiS

- Developed software for an autonomous humanoid robot (NAO, developed by Aldebaran) using Python.
- Deployed, validated and tested software on the robot.

- Developed internal tools, including a C# user interface and Python automation scripts.

https://cordis.europa.eu/project/id/687772

Education

September 2008 - November 2015

Aristotle University of Thessaloniki, *Diploma in Electrical and Computer Engineering* - Thessaloniki, GR

Grade: 7.45 / 10

Department: Electronics and Computer Engineering

Thesis title: Edge Detection Algorithms Acceleration on Zynq-7000

Supervisor: Loukas Petrou

Skills

Programming: C, C++, Go, Python, Bash, POSIX shell, Rust, Assembly (x86, AVR, PIC)
Build Systems: Make, CMake, Meson
Debugging & Profiling: GDB, Valgrind, ASan, Delve, strace, kprobes, eBPF, Linux kernel tracepoints, perf, gprof
Operating Systems: Linux, OpenBSD, Windows
Tools: Git, Wireshark, tcpdump, pfSense
Network Protocols: TCP/IP, UDP, SCTP, HTTP, DNS
Mobile Network Protocols: GTP-C, GTP-U, GTPv2-C, Diameter, S1AP, NGAP, NAS, PFCP
Virtualisation: KVM, QEMU, Cloud Hypervisor, Proxmox VE, Vagrant, libvirt
Containers: Docker
CI/CD & Automation: Gitlab CI, Jenkins, Ansible

Languages

Greek: Native
English: Advanced (C1)
German: Intermediate (B1)

Open Source Contributions

● Linux kernel: perf tool memory leak fixes (merged upstream)
https://lore.kernel.org/all/20260807-perf-kvm-leaks-v1-0-7b5e2f97905a@gmail.com/